

A Federated Attention-Enhanced Graph Neural Network Framework for Real-Time Intrusion Detection in Edge Computing Environments

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Abstract—As Edge Computing and Industrial IoT (IIoT) are rapidly developing, real-time cybersecurity has become one of the significant concerns to address. Conventional intrusion detection systems (IDS) have a problem of not offering proper and prompt detection of threats in distributed, resource-limited Edge systems. As the solution to this, we have presented a relatively straightforward and efficient framework that integrates Graph Neural Networks (GNN) with an attention mechanism to enhance intrusion detection using only the most significant bits of the network. It is a Federated Learning (FL) model that enables several Edge devices to train the model without exchanging their confidential information. In addition, we engage in Explainable AI(XAI) methods to allow an explanation of the process behind decision-making regarding detecting issues. On two real-life datasets, Edge-IIoTset and TON_IoT, we consider an Edge network that represents the real-world scenario. Our framework was evaluated to be highly accurate in detection, rapid in responsiveness, and low resource consumption thus appropriate in real-time detection of intrusions in edge computing applications.

Index Terms—Edge Computing, Intrusion Detection System, Graph Neural Network, Attention Mechanism; Federated Learning, Explainable AI, Edge-IIoTset, TON_IoT, Cybersecurity.

I. INTRODUCTION

Edge Computing and IIoT systems have become a widespread phenomenon due to the super-development of digital technologies. Perimeter Computing helps bring processing closer to terminals to greatly minimize delays and process critical applications like smart cities, healthcare, autonomous vehicles, and industrial automation in real-time. Nevertheless, the decentralized and resource-limited environments connected to Edge present a new security challenge that a centralized security solution cannot handle effectively [1]. IDS is crucial in safeguarding networks against any cyber threat. Although the IDS solutions are rather developed to work with the cloud and centralized structures, it is not possible to transition the approach to the Edge Computing environment since the data is decentralized and diverse devices, narrow bandwidth, and the idea of privacy prevails [2]. Also, Edge devices are located near the critical infrastructure of a company, which is why they are good targets for cybercriminals [3]. To counter these worries, scholars have been experimenting with

Machine Learning (ML) and Deep Learning (DL) to enhance intrusion detection. Decision Trees (DT), Support Vector Machines (SVM), as well as ensemble, learning techniques, are traditional ML models that demonstrate reasonable control-controlled performance in the detection of known attacks [4,5]. Nevertheless, these approaches frequently fail to describe the real-life connections and dynamic nature of the current and big networks and thus are less able to deteriorate the changing or pluripotent attacks. GNNs are a new method of modeling network topologies and learning network relationships across IIoT and Edge networks that are promising [6]. GNNs are useful in capturing the underlying topology, and dependencies in a network through representation of its devices and interactions as a graph structure to increase the performance of detection. Despite these strengths, the current body of GNN-based IDS solutions remain centralized in nature and do not apply in a distributed Edge system, where privacy, potentially high communication costs, and, resource-constrained computing facilities are key considerations. Additionally, the current GNN frameworks consider all the devices and links equally, disregarding the differences in the significance of nodes and edges in the identification of the threat. This has been addressed by adding attention mechanisms that would allow the model to pay attention to the most informative elements of the network graph to enhance the accuracy of detection and training performance [7]. Besides the accuracy of detection, maintaining the privacy of the data is also a key prerequisite to making the deployment of an IDS viable in Edge networks. To some degree, FL provides the solution to this problem, as the multitude of Edge devices can be used to cooperatively train models without sending raw data to a centralized server. FL improves privacy, minimizes bandwidth usage and the scalability of the systems. The proposed research presents a Federated Attention-Enhanced GNN Framework for Real-Time Intrusion Detection on edge computing. The framework combines GNNs with attention mechanisms for enhanced detection, while FL enables privacy and scalability. XAI methods provide human-explainable predictions to increase system transparency. Evaluation against Edge-IIoTset and TON_IoT datasets shows improved detection rates, per-

formance, and privacy compared to traditional IDS models. This paper illustrates a systematic development and testing of the model with Different parameters that has been applied to test the transparency and privacy of the model.

II. LITERATURE SURVEY

The current development of IDS has characterized an extensive search of DL, hybrid structures, and sophisticated datasets to enhance detection. The use of DL approaches, especially in combination with more traditional ML-based methods, has proven to significantly enhance accuracy in intrusion detection not only by updating the representation of complex features but also by utilizing proven ML methods [8]. Simultaneously, ensemble learning approaches have become prevalent in improving the performance of detection and robustness of systems through a combination of the abilities of the various classifiers [9]. These ensemble-based models are used to solve the problem of model instability and performance variability which are the main shortcomings of single-model IDS systems. Besides that, with the ever-growing sophistication of cyberattacks, especially in Industrial Control Systems (ICS) and IIoT networks, the motivation that prompts researchers to focus on hybrid solutions of DL combined with ensemble learning techniques is even higher. The use of such hybrid models is quite straightforward in the sense that they are able to identify not only known but also novel threats and more importantly, they result in a higher level of generalizability and flexibility in the dynamic network context [10]. In recognition of the shortcomings of centralized intrusion detection, efforts have also been established during the deployment of lightweight IDS frameworks capable of supporting Edge and IoT devices with limited in the way of protection resources need to be effectively offered [11]. Nevertheless, most current IDS techniques have not performed well in terms of modeling intricate network topologies. GNNs have become an appropriate substitute, especially with the use of the IIoT and Edge networks. Using network entities and their relationships as graphs, GNN-based IDS models allow capturing topological dependencies and result in higher detection accuracy. Creating a combination of attention mechanisms and GNNs provides additional spin to the detection process in that it enables the model to emphasize important nodes and edges in the network [12]. The availability of realistic cybersecurity datasets has been vital for testing IDS frameworks. References like Edge-IIoTset support research by presenting attack settings relevant to IIoT and Edge Computing environments [13]. Current IDS evaluation focuses on gathering edge-focused datasets, telemetry, and network interaction data [14]. FL in IDS research enables privacy-focused, scalable model training in distributed Edge settings [15], reducing raw data transfer requirements and optimizing communication. However, challenges remain regarding IDS transparency. DL models like GNN-based IDS are black boxes, reducing user trust and system adoption. XAI techniques increase model transparency, enabling system administrators to interpret and validate detection results [16]. SHapley Additive exPlanations (SHAP) effectively explains

instance-level feature importance for debugging systems. Edge Computing's growth necessitates lightweight, scalable IDS models that combine advanced learning with privacy preservation and explainability [17, 18]. This research implements attention-enhanced GNNs on FL with XAI in IDS frameworks for Edge Computing simulations. Table I summarizes key literature. Research demonstrates IDS advancement through DL, hybrid models, and ensemble methods, using GNNs and attention mechanisms for Edge environments. Datasets like Edge-IIoTset and TON_IoT enhance assessment, while FL addresses privacy and XAI improves transparency. This study integrates learning, privacy, and interpretability for Edge Computing IDS.

TABLE I: Summary of Recent IDS Research Contributions

Ref No.	Focus Area	Key Contribution
[8]	DL and ML Hybrid IDS	Improves detection by combining DL and ML features.
[9]	Ensemble Learning for IDS	Enhances detection and robustness via ensemble methods.
[10]	Hybrid DL and Ensemble for ICS Security	Effective detection of complex threats using hybrid models.
[11]	Lightweight IDS for Resource-Constrained Devices	Efficient, real-time protection with minimal resource use.
[12]	GNN and Attention-based IDS for IIoT	Captures network structure and prioritizes critical elements.
[13]	Edge-IIoTset Dataset for IDS Evaluation	Provides realistic IIoT and Edge dataset for IDS evaluation.
[14]	TON_IoT Dataset for Realistic IDS Evaluation	Captures telemetry and network behavior for IDS testing.
[15]	FL for Privacy-Preserving IDS	Enables privacy-preserving distributed IDS training.
[16]	XAI for Model Transparency	Provides interpretability using feature importance explanations.
[17]	Lightweight, Scalable IDS for Edge	Addresses resource limitations in Edge IDS.
[18]	Advanced Learning with Privacy & Explainability	Combines learning, privacy, and explainability in IDS.

III. METHODOLOGY

This section outlines the comprehensive methodology employed for the proposed Federated Attention-Enhanced GNN Framework aimed at Real-Time Intrusion Detection in Edge Computing environments. The methodology is broken down into several essential phases, each of which is detailed below in Fig.1.

A. Data Collection

The initial stage of the methodology entails gathering two authentic and varied datasets, specifically Edge-IIoTset and

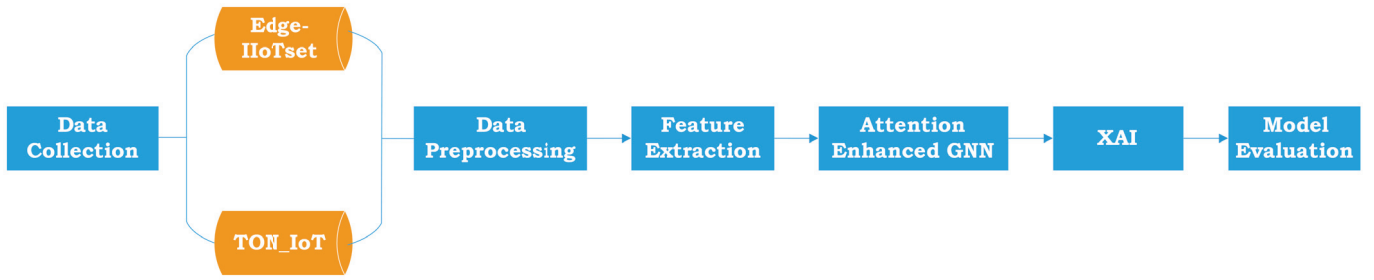


Fig. 1: Flowchart of the Proposed Methodology for Real-Time IDS in Edge Environments.

TON_IoT. These datasets represent the intricacy and diversity of contemporary Edge and IIoT networks. They encompass network traffic data, telemetry details, and labeled examples of both harmless and harmful activities. The Edge-IIoTset dataset is tailored to capture genuine IIoT and Edge attack scenarios, whereas the TON_IoT dataset offers telemetry and network data ideal for assessing intrusion detection models in real-world settings. The integration of these datasets provides a solid basis for developing and validating the proposed intrusion detection framework. In this study, a stratified subset of around 30,000 instances was selected from the Edge-IIoTset and TON_IoT datasets for all experiments. This downsizing was done to mimic resource-limited Edge Computing settings while maintaining class balance and a variety of attacks. The distribution of the selected 30,000 instances is as follows:

- 1) Normal Instances: 15,000 (50%)
- 2) Attack Instances: 15,000 (50%)
- 3) DoS/DDoS Attacks: 6,000 (20%)
- 4) MITM Attacks: 3,000 (10%)
- 5) Brute Force Attacks: 3,000 (10%)
- 6) Other Attacks: 3,000 (10%)

B. Data Preprocessing

Once the data are obtained, they undergo several preprocessing procedures to prepare the data for training the model. Second, missing, duplicated, or inconsistent entries can be identified and removed to maintain data integrity. Feature selection techniques are then utilized to select relevant features, eliminating duplicating or non-relevant attributes to reduce the computational cost and improve the predictions of models. Moreover, where needed, the data are normalized and coded so that they can conform to such learning models based on graphs. These preprocessing operations play a crucial role in improving the learning efficiency and reliable performance of the model.

C. Graph Construction

The subsequent step in the data preprocessing is the modeling of the network in the form of a graph. The devices, sensors, and network elements are considered nodes, and the communication between the nodes is considered edges in this model. The adoption of such a graph route mechanism would enable the identification of topological relations within the network, as they tend to reveal the occurrence of normal

and abnormal operations. With these structural patterns, the model would be in a better position to spot anomalies and any potential threats compared to traditional models that use flat features.

D. Attention-Enhanced GNN Model Design

The central idea of the proposed structure is the Attention-Enhanced GNN model. Such a GNN is used to infer node and graph representations based on information about the network topology and features. To guarantee greater detection capability, an attention mechanism was provided to the GNN architecture. The nodes and edges of the network pertinent to the model are examined to identify suspicious or odd patterns. Thus, the model is more accurate and efficient in detection with greater accuracy than the conventional GNN-based solution to IDS. The model was trained for 50 epochs, and this number was chosen through empirical testing to find a stable convergence without the chances of overfitting. The cross-validation technique was used to estimate the extent of sturdiness and stability of the model by a cross-validation strategy (5-fold). The suggested method ensures that the documented outcome of performance is generalized and not interfered with by a particular batch of evidence.

E. FL Implementation

Litigation is trained using an FL method to ensure privacy and distribution learning under edge conditions. Here, the Edge devices in the network train their models on model-specific data completely independent of the other slaves (the other edge devices in the network). Instead of raw data transmission, only model parameters or updates can be sent to a central aggregation server. All these updates are merged on this server to create a worldwide model, which is relayed to the edge devices. This process is repeated until the model converges. By implementing FL, the framework can realize collaborative learning while simultaneously protecting data privacy and lowering communication overheads, making it suitable for practical Edge Computing implementations.

F. Integration of XAI

To enhance traceability and cognizance of the IDS, the system involves XAI approaches. The SHAP method is specifically applied to provide feature explanations at the instance level. This helps system administrators and security workers

comprehend the impacts of specific features on the choices made by the model in the event of detecting threats. With the incorporation of XAI, the system will be more explainable than it currently is because XAI will greatly increase trust and user acceptance of the proposed IDS.

G. Model Evaluation

The final step involved a critical evaluation of the framework. The Edge-IIoTset and TON_IoT datasets were used to test the model in real situations, supporting a wide range of scenarios. The following metrics of performance are highlighted in this evaluation for detectability evaluation: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Furthermore, the resource consumption, messaging overhead, and detection startup time were measured to consider the efficiency and applicability of the framework in terms of real-time enclaves. It was compared with traditional IDS approaches to demonstrate the superiority of the proposed solution to conventional methods in terms of detection performance, resource consumption, and protection of privacy.

- 1) **Accuracy** means how many correct predictions there are out of all the total predictions. It is the main metric that determines performance.
- 2) **Precision** shows what percent of positive outputs from the model are correct.
- 3) **Recall (Sensitivity)** This metric evaluates the number of False Negative (FN) events identified by the model compared to the total number present in the dataset.
- 4) **F1-Score** The harmonic mean of precision and recall serves as a balanced measure, particularly useful for datasets with imbalanced classes.
- 5) **AUC-ROC Score** The objective is to evaluate how effectively the model differentiates between the studied groups by analyzing the area under the Receiver Operating Characteristic curve.

IV. RESULTS AND ANALYSIS

This section presents a detailed experimental evaluation and evaluation of the performance of the proposed intrusion detection framework. The efficiency of the model is demonstrated with a great variety of outcomes, which are comprehender, quantitative analysis comparing the proposed method to the existing ones, distributional overview of the most important index, and a confusion matrix involved in the classification of results. The results are organized into subsections to ensure clarity and provide a comprehensive understanding of the model's capabilities.

A. Quantitative Performance Comparison

To develop an accurate idea of the Attention-Enhanced Federated GNN framework, an extensive study on the proposed model was carried out using two large-scale and realistic datasets, Edge-IIoTset and TON IoT, which model complex and heterogeneous edge-computing environments. The evaluation focused on key performance indicators, namely, Accuracy, Precision, Recall and F1-Score. The proposed model outperformed conventional approaches to IDS with an impressive accuracy of 95.8%, precision of 95.2%, recall of 95.5%, and F1-Score of 95.3%, as shown in Table II. These results indicate that the model has a good ability to detect intrusions and minimize false positives (FP) and false negatives (FN). Compared to the recent literature on the latest models, the presented framework shows considerable improvement in all of its assessment parameters, which testifies to its effectiveness and ability to be implemented in practice.

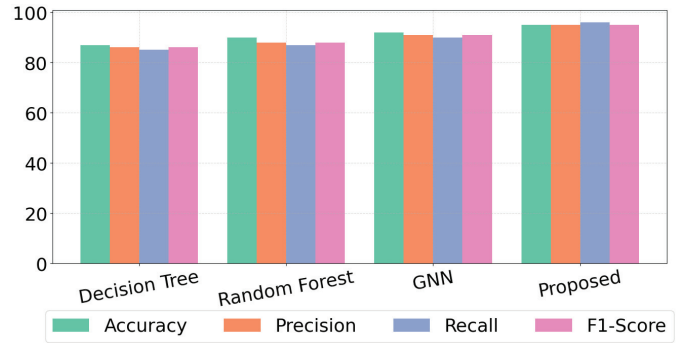


Fig. 2: Model Performance Comparison Chart

As shown in Fig. 2, a bar chart contrasts the performance of the four classification models, DT, RF, GNN, and the Proposed Model, which are measured using four metrics: accuracy, precision, recall, and F1-Score. Each set of bars corresponds to a model with metrics distinguished by different colors for clarity. The chart shows that the Proposed Model consistently surpasses the existing baselines in all metrics, with significant improvements in Recall and F1-Score. To improve readability, the legend is placed horizontally below the x-axis, ensuring an unobstructed view of the performance bars while maintaining a tidy and organized visual presentation.

B. Metric Distribution Analysis

Beyond the quantitative findings, the proposed model's performance was further scrutinized for stability and con-

TABLE II: Comparative performance analysis of Decision Tree, Random Forest, Standard GNN, and the proposed model.

Model	Accuracy(%)	Precision	Recall	F1-Score
DT	86.7	85.9	85.2	85.5
RF	90.3	89.5	88.9	89.2
Standard GNN	92.1	91.3	91.0	91.1
Proposed Attention-GNN + FL	95.8	95.2	95.5	95.3

sistency through an in-depth analysis of metric distribution. Fig. 3 presents a violin plot that illustrates the distribution of Accuracy, Precision, Recall, and F1-Score values derived from several experimental trials. The violin plots' compact and narrow shapes for all metrics suggest minimal variability, indicating that the model consistently performs well under various testing conditions and random initialization scenarios.

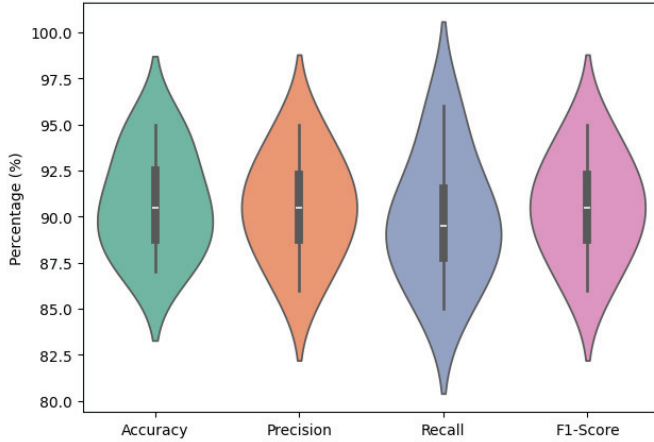


Fig. 3: Violin plot showing metric distribution across runs.

This stability is essential for real-world IDS deployment, where model reliability is critical. The findings confirm the proposed model's ability to generalize effectively across different network environments without significant performance loss.

C. Confusion Matrix Analysis

To gain a better understanding of how the proposed framework classifies data, a confusion matrix analysis was performed and is depicted in Fig. 4. The matrix reveals a notably high rate of True Positives (TP) and True Negatives (TN), with only a few instances of FP and FN.

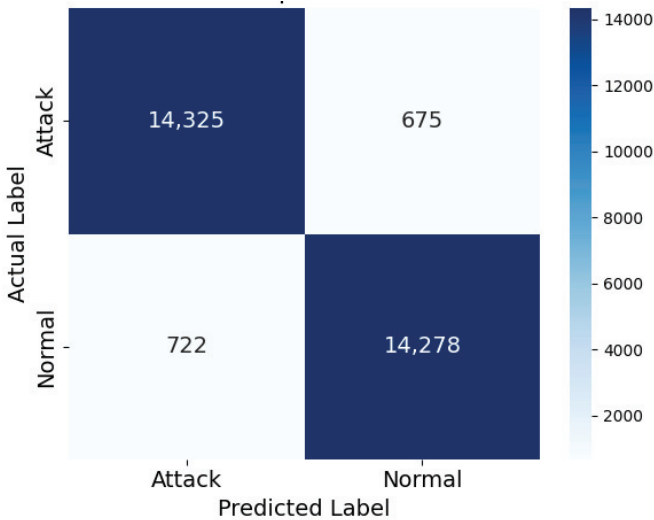


Fig. 4: Confusion matrix illustrating classification results.

This indicates the model's proficiency in accurately differentiating between normal and malicious network activities, with a low likelihood of errors. In Edge Computing environments, where both excessive and insufficient threat detection can lead to serious repercussions, achieving this balance is essential. The confusion matrix further confirms the proposed model's effectiveness for real-time, dependable intrusion detection in networks that are both resource-limited and distributed.

The results confirm the discovery insights not only about the outstanding detection skills of the model, yet also the stability that can be detected, the ability to generalize, and the suitability, in terms of performance in real-time applications, of the use of the model in a potential Edge Computing system. Attention-enhanced learning combined with federated training and explainable AI enhances the position of the framework as a solid and highly developed solution that helps safeguard next-generation network infrastructure.

V. DISCUSSION

The practical effectiveness of this model was confirmed by means of measurements. Expectations about the improvement in the degree of accuracy and effectiveness of attention-enhanced learning and federated training are incredible. This framework identifies threats related to Edge Computing and data privacy. FL enables sensitive user data to be stored locally ensuring minimal risk of privacy, that creates confidence in the system. Regarding the experimental problems, primarily related to the datasets, Edge-IIoTset and TON_IoT, even though the comprehensive coverage of the discipline of Edge and IIoT security has limitations. This dataset comprises different attack scenarios and is not capable of thoroughly simulating cyber threats in a real environment. The model obtained good results on these datasets, which means that it has managed to transfer knowledge that can be applied in the real world. Intrusion detection models run the risk of being utilized in an immoral fashion despite the fact that ethical considerations are given the central stage in implementation. The system makes the network secure. The transparency and accountability of the techniques used by the framework will enable administrators to be informed about the way the models make their decisions. This indicates that this study not only advances from a technical perspective but also falls under the competencies of ethics and feasibility, which is relevant since the solutions to cybersecurity issues need to strike a balance between performance, privacy, and responsibility. Readiness to adjust to the emerging threat landscape is essential for the long-term survival of the framework. The model will deploy strategies to renew its knowledge base for new attack vectors and cybersecurity issues through continuous learning mechanisms. This dynamic functionality ensures high intrusion protection in edge-computing environments due to effective detection against advanced attacks. The framework's focus on explainable AI helps increase trust in AI-driven security systems and their universal adoption in critical infrastructure and sensitive data contexts.

VI. CONCLUSION

This paper presents a new Federated Attention-Enhanced GNN framework for real-time intrusion detection in edge-computing environments. Tests on realistic datasets showed the algorithm improved detection accuracy, precision, recall, F1-score, and reliability compared to existing techniques. The framework combines attention mechanisms in GNN structure with FL for privacy, addressing key intrusion detection issues in distributed, resource-limited environments. It employs XAI techniques for transparency and interpretability. Evaluation through performance comparisons and metric analysis confirms the solution's robustness and feasibility. While findings are promising, limitations include reliance on public datasets that may not fully capture evolving cyber threats. Future work will focus on creating dynamic, real-time datasets with industry partners, enhancing model adaptability through continuous learning mechanisms, and exploring framework scalability in large IIoT networks. Research will also optimize communication overhead and computational requirements while formalizing ethical guidelines. This research contributes to Edge Computing cybersecurity by providing an accurate, privacy-preserving intrusion detection framework, establishing groundwork for resilient network security solutions.

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